AHARADA EDUCATION

Excellence in Aviation, Aerospace & Advanced Computing Technologies

OFFICIAL CURRICULUM STUDY MATERIAL & LECTURE NOTES

COURSE & SUBJECT SPECIFICATION

Course: B.Sc Aeronautical Science
Subject Code: MBA - AI105
Subject Name: Data Base Management & SQL
Semester: Semester 5
Target Unit: Unit 2
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Unit 2: Data Base Management & SQL

Subject: Data Base Management & SQL (MBA - AI105)

1. Unit Syllabus & Module Breakdown

Entity Relationship (ER) Model; ER Diagrams; Entities, Attributes, Relationships; Cardinality; Keys (Primary, Foreign, Composite, Candidate); Relational Schema; Functional Dependencies; Database Design Process; Normalization (1NF, 2NF, 3NF, BCNF); Denormalization; Data Integrity Constraints; Business database modelling.

2. Core Learning Objectives & Key Concepts

- **Topic 2.1:** Entity Relationship (ER) Model
- **Topic 2.2:** ER Diagrams
- **Topic 2.3:** Entities, Attributes, Relationships
- **Topic 2.4:** Cardinality
- **Topic 2.5:** Keys (Primary, Foreign, Composite, Candidate)
- **Topic 2.6:** Relational Schema
- **Topic 2.7:** Functional Dependencies
- **Topic 2.8:** Database Design Process
- **Topic 2.9:** Normalization (1NF, 2NF, 3NF, BCNF)
- **Topic 2.10:** Denormalization
- **Topic 2.11:** Data Integrity Constraints
- **Topic 2.12:** Business database modelling.

3. Detailed Chapter Notes & Concepts

3.1 Entity-Relationship (ER) Modeling & Diagrams

The ER model defines the conceptual view of a database. Key components include Entities (real-world objects with independent existence), Attributes (properties or characteristics of entities such as simple, composite, single-valued, multi-valued, and derived), and Relationships (associations among entities). Cardinality ratios define the number of relationship instances (1:1, 1:N, M:N). Participation constraints specify mandatory or optional existence.

3.2 Relational Schema & Relational Integrity Constraints

A relational database consists of tables (relations) with rows (tuples) and columns (attributes). Integrity constraints maintain data validity:

• Entity Integrity: The primary key cannot contain null values.
• Referential Integrity: A foreign key value must match an existing primary key or be null.
• Domain Integrity: Column values must adhere to defined data types and ranges.

3.3 Normalization & Functional Dependencies

Normalization minimizes data redundancy and avoids insertion, update, and deletion anomalies:

• 1NF (First Normal Form): Eliminates repeating groups; all attribute values are atomic.

• 2NF (Second Normal Form): In 1NF and contains no partial functional dependencies (every non-prime attribute is fully functionally dependent on the entire primary key).

• 3NF (Third Normal Form): In 2NF and contains no transitive functional dependencies (X -> Y and Y -> Z where Z is non-prime).

• BCNF (Boyce-Codd Normal Form): Stricter version of 3NF where for every functional dependency X -> Y, X must be a super key.

4. Review Questions & Self-Assessment

1. Explain the fundamental architectural distinctions between physical and logical data independence.
2. Discuss how the topics covered in this unit apply to industrial workflows in B.Sc Aeronautical Science.
3. Formulate standard case diagrams and explain key constraints discussed under Unit 2: Data Base Management & SQL.
4. Review previous university examination questions related to Data Base Management & SQL (MBA - AI105).